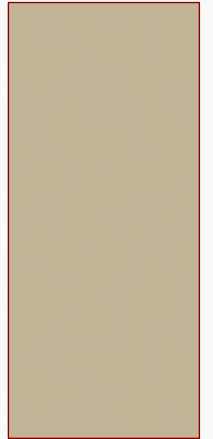


CAMERAS

ADVANCED GAME DESIGN & DEVELOPMENT

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CAMERAS

- Cameras are the devices that capture and display the world to the player.
- You can have an unlimited number of cameras.
- They can be set to render in any order, at any place on the screen, or only certain parts

CAMERA PROPERTIES

- **Clear Flags**
 - Determines which parts of the screen will be cleared. This is handy when using multiple Cameras to draw different game elements.
- **Background**
 - Color applied to the remaining screen after all elements in view have been drawn and there is no skybox.
- **Culling Mask**
 - Include or omit layers of objects to be rendered by the Camera. Assign layers to your objects in the Inspector
- **Projection**
 - **Perspective** - Camera will render objects with perspective intact.
 - **Orthographic** - Camera will render objects uniformly, with no sense of perspective.
- **Size**
 - (when Orthographic is selected)The viewport size of the Camera when set to Orthographic.
- **Field of view**
 - (when Perspective is selected)Width of the Camera's view angle, measured in degrees along the local Y axis.
- **Clipping Planes**
 - **Near**
 - The closest point relative to the camera that drawing will occur.
 - **Far**
 - The furthest point relative to the camera that drawing will occur.



CAMERAS PROPERTIES

- **Normalized View Port Rect**
 - Four values that indicate where on the screen this camera view will be drawn, in Screen Coordinates (values 0-1).
 - **X**
 - The beginning horizontal position that the camera view will be drawn.
 - **Y**
 - The beginning vertical position that the camera view will be drawn.
 - **W**
 - (Width)Width of the camera output on the screen.
 - **H**
 - (Height)Height of the camera output on the screen.
- **Depth**
 - The camera's position in the draw order. Cameras with a larger value will be drawn on top of cameras with a smaller value.
- **Rendering Path**
 - Options for defining what rendering methods will be used by the camera.
 - **Use Player Settings**
 - This camera will use whichever Rendering Path is set in the Player Settings.
 - **Vertex Lit**
 - All objects rendered by this camera will be rendered as Vertex-Lit objects.
 - **Forward**
 - All objects will be rendered with one pass per material.
- **Target Texture**
 - (Unity Pro only)Reference to a [Render Texture](#) that will contain the output of the Camera view. Making this reference will disable this Camera's capability to render to the screen.
- **HDR** Enables High Dynamic Range rendering for this camera.



CAMERA HINTS

- **Hints**

- Cameras can be instantiated, parented, and scripted just like any other GameObject.
- To increase the sense of speed in a racing game, use a high **Field of View**.
- Cameras can be used in physics simulation if you add a **Rigidbody** Component.
- There is no limit to the number of Cameras you can have in your scenes.
- Orthographic cameras are great for making 3D user interfaces

CAMERA HINTS

- **Hints**

- If you are experiencing depth artifacts (surfaces close to each other flickering), try setting **Near Plane** to as large as possible.
- Cameras cannot render to the Game Screen and a Render Texture at the same time, only one or the other.
- Pro license holders have the option of rendering a Camera's view to a texture, called Render-to-Texture, for even more unique effects.
- Unity comes with pre-installed Camera scripts, found in **Components->Camera Control**. Experiment with them to get a taste of what's possible.

CAMERA TUTORIALS

- Great Tutorials

- <http://unity3d.com/learn/tutorials/projects/stealth/camera-movement>
- <http://unity3d.com/learn/tutorials/modules/beginner/live-training-archive/cameras>
- <http://unity3d.com/learn/tutorials/projects/roll-a-ball/moving-the-camera>
- <http://unity3d.com/learn/tutorials/projects/space-shooter/camera-and-lighting>